

Probability

Statistics I • Section 3

Zong-Wei Yeh

Department of Finance, National Chung Cheng University

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Outline

- 3.1 Random experiments, events, and the axioms
- 3.2 Classical probability and counting
- 3.3 Conditional probability, independence, multiplication rule
- 3.4 Total probability and Bayes' theorem

Random Experiments, Events, and the Axioms

Why we need probability

- We sample in order to speak about a population we cannot see. Every such statement can be wrong.
- To **measure** that chance we need probability. We build it from a few axioms.
- Sections 4 to 7 all rest on this section: random variables, distributions, sampling distributions.

Experiments, sample space, events

Definition 3.1 (Random experiment, sample space, event)

1. A *random experiment* (隨機實驗) has two or more possible outcomes. Which one occurs is not known in advance.
 2. The *sample space* (樣本空間) S is the set of all basic outcomes, defined so that exactly one occurs.
 3. An *event* (事件) is any subset $A \subseteq S$; it occurs if the outcome that happens belongs to A .
- The experiment is the situation itself: a die roll, tomorrow's market move.
 - An event is a question about the outcome, such as “was it even?”.

What a sample space looks like

Experiment	Sample space S	One event, as a set
Toss a coin once	$\{H, T\}$	head: $\{H\}$
Toss a coin twice	$\{HH, HT, TH, TT\}$	at least one head: $\{HH, HT, TH\}$
Roll a die	$\{1, \dots, 6\}$	even: $\{2, 4, 6\}$
Count wins in 5 bids	$\{0, 1, \dots, 5\}$	at least 4 wins: $\{4, 5\}$
A stock over two days	$\{UU, UN, NU, NN\}$	rises on day 1: $\{UU, UN\}$

- H : head. T : tail. U : the price rises that day. N : it does not rise.
- If the outcome is UN , “rises on day 1” occurs. “Never rises” = $\{NN\}$ does not.
- Two dice: record the pair (36 outcomes) or the sum ($\{2, \dots, 12\}$). With fair dice, only the pairs are equally likely.

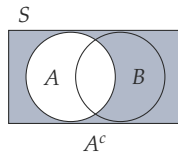
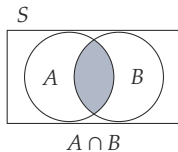
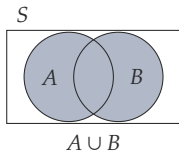
Combining events

Definition 3.2 (Set operations)

For events $A, B \subseteq S$: $A \cup B$ (*union*: either or both), $A \cap B$ (*intersection*: both), A^c (*complement*: not A). A, B are *mutually exclusive* (互斥) if $A \cap B = \emptyset$, the *empty event* (空事件). A list of events (finite or not) is mutually exclusive if every pair is. $E_1, \dots, E_k$ are *collectively exhaustive* (周延) if $E_1 \cup \dots \cup E_k = S$.

- A list that is both mutually exclusive and collectively exhaustive is a **partition** of S . We need it in §3.4.

Set operations on a Venn diagram



- Shading shows which outcomes each operation keeps.

Combining events: two trading days

$S = \{UU, UN, NU, NN\}$.

A : rises on day 1, $A = \{UU, UN\}$. B : rises on day 2, $B = \{UU, NU\}$.

- $A \cup B = \{UU, UN, NU\}$: rises on at least one day.
- $A \cap B = \{UU\}$: rises on both days.
- $A^c = \{NU, NN\}$: does not rise on day 1.
- $C = \{NN\}$ (never rises): $A \cap C = \emptyset$, so A and C are mutually exclusive.
- $\{UU\}, \{UN, NU\}, \{NN\}$ (rises on 2, 1, 0 days) are mutually exclusive, with union S . So they form a partition.

The axioms of probability

Definition 3.3 (Axioms)

A probability assigns $P(A)$ to each event $A \subseteq S$ so that these three axioms (機率公理) hold:

(A1) $P(A) \geq 0$;

(A2) $P(S) = \underline{1}$;

(A3) for mutually exclusive $A_1, A_2, \dots$: $P(\bigcup_i A_i) = \sum_i P(A_i)$.

- We use (A3) more than the other two. It is the rule behind almost every calculation in this section.
- (A1) and (A2) only fix the scale. Everything else is a **theorem**.

Example 3.4

Example 3.4

Two trading days, $S = \{UU, UN, NU, NN\}$, with $P(\{UU\}) = 0.3$, $P(\{UN\}) = 0.2$, $P(\{NU\}) = 0.2$, $P(\{NN\}) = 0.3$. (a) Use the axioms to find $P(\text{rises on day 1})$ and $P(\text{rises on at least one day})$. (b) Why can 0.4, 0.3, 0.2, 0.2 not be these four probabilities?

The complement rule

Proposition 3.5 (Complement rule)

$P(A^c) = 1 - P(A)$; hence $P(\emptyset) = 0$ and $0 \leq P(A) \leq 1$.

- A and A^c are mutually exclusive, and together they are all of S .
- So (A3) and (A2) give $P(A) + P(A^c) = 1$.

The addition rule

Proposition 3.6 (Addition rule)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- $P(A) + P(B)$ counts the overlap twice, so we subtract it once.
- If A, B are mutually exclusive, the rule reduces to $P(A \cup B) = P(A) + P(B)$.

Example 3.7

Example 3.7

Roll one fair die, $S = \{1, \dots, 6\}$, $A = \{\text{even}\}$, $B = \{1, 2, 3\}$. Find $A \cup B$, $A \cap B$, A^c and their probabilities.

Example 3.8

Example 3.8

Investors: $P(\text{stocks}) = 0.5$, $P(\text{bonds}) = 0.4$, $P(\text{both}) = 0.3$. Find $P(\text{stocks or bonds})$ and $P(\text{neither})$.

An upper bound for a union

Proposition 3.9 (Boole's inequality)

For any events $A_1, \dots, A_n$, mutually exclusive or not,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

- **Boole's inequality** (布爾不等式) bounds $P(\text{at least one occurs})$.
- It needs no information about the overlaps. The sum counts each overlap more than once.
- For mutually exclusive events the bound is exact: that is (A3).

Classical Probability and Counting

When every outcome is equally likely

Definition 3.10 (Classical probability)

If S has N equally likely outcomes and A contains N_A of them, the *classical probability* (古典機率) of A is

$$P(A) = \frac{N_A}{N}.$$

- Computing $P(A)$ is then a pure counting problem.
- So we need ways to count without listing.

Example 3.11

Example 3.11

Roll two fair dice, one red and one blue. (a) Find $P(\text{sum} = 7)$. (b) A student uses $S = \{2, 3, \dots, 12\}$, which has 11 outcomes, and answers $1/11$. What is wrong?

Counting without listing

- **The multiplication principle** (乘法原理). A task is done in stages. There are n_1 choices at the first stage, n_2 at the second, and so on up to n_k . The number of choices at each stage does not depend on what was chosen earlier. The number of ways is

$$n_1 \times n_2 \times \cdots \times n_k.$$

- The other two counts are built on this one.

Two ways to choose x from n

Definition 3.12 (Permutations and combinations)

From n distinct objects, choosing x (here $n! = n(n-1) \cdots 2 \cdot 1$, with $0! = 1$):

$$\text{ordered (permutation (排列))}: P(n, x) = \frac{n!}{(n-x)!}$$

$$\text{unordered (combination (組合))}: C(n, x) = \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

- A permutation counts arrangements. A combination counts sets.

Deciding which count to use

- Every time, we must decide whether order matters. A ranking is a permutation; a committee is a combination.
- Linked by $C(n, x) = P(n, x) / \underline{x!}$, since each set of x objects can be arranged in that many ways.
- $C(n, x)$ returns in Section 5, counting the binomial's success patterns.

Example 3.13

Example 3.13

A password is 2 letters (A–Z) then 3 digits (0–9), repeats allowed. How many passwords?

Example 3.14

Example 3.14

From 5 analysts: (a) how many ordered top-3 rankings? (b) how many 3-person committees?

Example 3.15

Example 3.15

A committee of 3 is chosen at random from 6 men and 4 women. Find $P(\text{exactly 2 women})$.

Conditional Probability, Independence, Multiplication Rule

Probability once something is known

Definition 3.16 (Conditional probability)

For $P(B) > 0$, the *conditional probability* (條件機率) of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

- Knowing B occurred shrinks the sample space from S to B .
- The conditional probability then measures how much of B is also in A .

Example 3.17

Example 3.17

Roll one fair die. Let $A = \{4, 5, 6\}$ and $B = \text{"the outcome is even"}$. Find $P(A)$ and $P(A \mid B)$. Why do they differ?

The probability that both happen

Proposition 3.18 (Multiplication rule)

For events with positive probability,

$$P(A \cap B) = P(A \mid B) P(B) = P(B \mid A) P(A).$$

- The **multiplication rule** (乘法法則) builds a sequence one stage at a time.
- That makes it ideal for draws without replacement.
- It is just the definition of conditional probability, rearranged.

When knowing one tells us nothing

Definition 3.19 (Independence)

A and B are *independent* (獨立) if $P(A \cap B) = P(A) P(B)$, and equivalently, when $P(B) > 0$, if $P(A \mid B) = P(A)$.

- So when the two events are independent, conditioning changes nothing.
- Independence is something to be **checked**, not taken for granted.
- Section 7 assumes independent samples. So this check matters there.

Example 3.20

Example 3.20

Of 100 investors (A = holds stocks, B = holds bonds), 30 are in $A \cap B$, 20 in $A \cap B^c$, 10 in $A^c \cap B$ and 40 in $A^c \cap B^c$. Find $P(B \mid A)$, $P(A \mid B)$; are A, B independent?

Example 3.21

Example 3.21

Two cards are drawn without replacement from a standard 52-card deck. Find $P(\text{both aces})$.

Example 3.22

Example 3.22

Two bulbs, each failing with probability 0.1, behaving independently. (a) The bulbs are wired in *series*. The circuit lights only if both bulbs work. Find the probability that it lights. (b) The bulbs are wired in *parallel*. The circuit lights if at least one bulb works. Find the probability that it lights. (c) Which wiring is more reliable, and why?

Total Probability and Bayes' Theorem

Averaging over cases

Theorem 3.23 (Total probability)

Let $B_1, \dots, B_k$ be mutually exclusive and collectively exhaustive, each with $P(B_i) > 0$. Then for any event A ,

$$P(A) = \sum_{i=1}^k P(A \mid B_i) P(B_i).$$

- A partition divides S into cases. Each outcome falls in exactly one case.
- **Total probability** (全機率) answers the forward question. It gives the overall probability of A from what we know inside each case.

Inverting the conditioning

Theorem 3.24 (Bayes' theorem)

Let $B_1, \dots, B_k$ be mutually exclusive and collectively exhaustive, each with $P(B_i) > 0$, and let $P(A) > 0$. Then

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_{i=1}^k P(A | B_i) P(B_i)}.$$

What Bayes' theorem is for

- We can measure $P(\text{effect} \mid \text{cause})$. We want $P(\text{cause} \mid \text{effect})$. **Bayes' theorem** (貝氏定理) turns the first into the second.
- The prior probability $P(B_j)$ is called the base rate (基本率). Most people forget it (see Example 3.26).

Example 3.25

Example 3.25

Factory 1 makes 60% of chips at a 2% defect rate. Factory 2 makes the other 40% at a 5% defect rate. (a) $P(\text{defective})$? (b) $P(\text{factory 2} \mid \text{defective})$?

Example 3.26

Example 3.26

A disease affects 1% of a population. A test is 99% sensitive (positive if diseased) with a 5% false-positive rate (positive though healthy). A randomly chosen person tests positive. Find $P(\text{diseased} \mid \text{positive})$.

Example 3.27

Example 3.27

A survey asks whether the respondent has ever cheated on an exam. To protect privacy, each respondent tosses a fair coin in private. On heads, the respondent tosses a second fair coin. The answer is “yes” if the second coin shows heads, and “no” if it shows tails. On tails, the respondent answers the cheating question truthfully. The tosses do not depend on whether the respondent has cheated. Let π be the proportion who have cheated. (a) Find $P(\text{yes})$ in terms of π . (b) 40% answer “yes”: estimate π . (c) After a “yes”, find the probability that the respondent has cheated.